Homework 2

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February 6, 2025
PY 355 Section A1

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Problem 1

1. Exponent rule
$$\int_0^1 a^y \,\mathrm{d}y = \left.\frac{a^y}{\ln a}\right|_0^1 = \frac{a-1}{\ln a}.$$
2. U-substitution, $u = ay$, $du = adx$, $v = -ay$, $dv = -vdx$.
$$\int_0^x \cosh y \,\mathrm{d}y = \int_0^x \frac{e^{ax} + e^{-ax}}{2} \,\mathrm{d}y = \frac{1}{2}\left(\int_0^x e^{ay} \,\mathrm{d}y + \int_0^x e^{-ay} \,\mathrm{d}y\right)$$
$$= \frac{1}{2}\left(\int_0^{ax} e^u \,\mathrm{d}u - \int_0^{-ax} e^v \,\mathrm{d}v\right).$$
3. Derivative of $\ln x$, u substitution $u = ay + b$, $du = 1/a \, dx$.
$$\int_0^x \frac{1}{ay+b} \,\mathrm{d}y = \int_0^x (ax+b)^{-1} \,\mathrm{d}y = \int_0^x \frac{1}{a} u^{-1} \,\mathrm{d}u = \frac{1}{a} \ln(ay+b)\bigg|_0^x = \frac{1}{a}\left(\ln(ax+b) - \ln b\right).$$
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